LLAIsys实验报告

cpu 推理优化：

原实现遍历MNK三重循环，没有用simd并行处理float的能力，且浮点数转换开销高，改进版本实现了f32 GEMV，nk>1用OpenMP并行，SIMD优化采用适配笔记本的AVX2，每轮可处理32个float,同时采用GEMM分块，速度明显快于三重循环;在第一次f16转float保存pack权重，后续直接调用不需重复转换。

```
(base) zhaocina@zhaocina-OMEN-by-HP-Gaming-Laptop-16-wf0xxx:~/Downloads/LLM/vllm/llaisys$ LLAISYS_LINEAR_USE_LEGACY=1 python test/ops/linear.py --pro
file
Testing Ops.linear on cpu
   decode-small-bias: out (1, 4096), x (1, 4096), w (4096, 4096), bias True, dtype <f32>
        Torch time: 0.89382 ms
        LLAISYS time: 8.09541 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 9.05713x
   decode-small-bias: out (1, 4096), x (1, 4096), w (4096, 4096), bias True, dtype <f16>
        Torch time: 0.57838 ms
        LLAISYS time: 35.56245 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 61.48634x
   decode-small-bias: out (1, 4096), x (1, 4096), w (4096, 4096), bias True, dtype <bf16>
        Torch time: 0.39344 ms
        LLAISYS time: 30.20403 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 76.76942x
   decode-small-nobias: out (1, 4096), x (1, 4096), w (4096, 4096), bias False, dtype <f32>
        Torch time: 0.93676 ms
        LLAISYS time: 8.03221 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 8.57443x
   decode-small-nobias: out (1, 4096), x (1, 4096), w (4096, 4096), bias False, dtype <f16>
        Torch time: 0.47022 ms
        LLAISYS time: 36.52745 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 77.68183x
   decode-small-nobias: out (1, 4096), x (1, 4096), w (4096, 4096), bias False, dtype <bf16>
        Torch time: 0.41016 ms
        LLAISYS time: 30.20095 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 73.63193x
```

```
(base) zhaocina@zhaocina-OMEN-by-HP-Gaming-Laptop-16-wf0xxx:~/Downloads/LLM/vllm/llaisys$ LLAISYS_LINEAR_USE_LEGACY=0 python test/ops/linear.py --pro
file
Testing Ops.linear on cpu
   decode-small-bias: out (1, 4096), x (1, 4096), w (4096, 4096), bias True, dtype <f32>
        Torch time: 1.26622 ms
        LLAISYS time: 1.46564 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 1.15749x
   decode-small-bias: out (1, 4096), x (1, 4096), w (4096, 4096), bias True, dtype <f16>
        Torch time: 0.58835 ms
        LLAISYS time: 1.20496 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 2.04803x
   decode-small-bias: out (1, 4096), x (1, 4096), w (4096, 4096), bias True, dtype <bf16>
        Torch time: 0.43894 ms
        LLAISYS time: 1.09565 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 2.49611x
   decode-small-nobias: out (1, 4096), x (1, 4096), w (4096, 4096), bias False, dtype <f32>
        Torch time: 1.12232 ms
        LLAISYS time: 1.34778 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 1.20088x
   decode-small-nobias: out (1, 4096), x (1, 4096), w (4096, 4096), bias False, dtype <f16>
        Torch time: 0.50013 ms
        LLAISYS time: 1.14851 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 2.29645x
   decode-small-nobias: out (1, 4096), x (1, 4096), w (4096, 4096), bias False, dtype <bf16>
        Torch time: 0.36925 ms
        LLAISYS time: 1.31569 ms
        speedup_vs_old: N/A (record baseline before rebuild)
        ratio_vs_torch: 3.56315x
```

由测试结果可得f32decode 已经非常接近 Torch，f16 / bf16大矩阵结果已经优于 Torch，各项较原始实现均有明显提速

nvidia适配CUDA:

实现的 Runtime API 包括：

- `get_device_count`

- `set_device`
- `device_synchronize`
- `create_stream`
- `destroy_stream`
- `stream_synchronize`
- `malloc_device`
- `free_device`
- `malloc_host`
- `free_host`
- `memcpy_sync`
- `memcpy_async`

  以及LLM的算子add,embedding,rms_norm,rope,swiglu,argmax,selfattention等

```
(base) zhaocina@zhaocina-OMEN-by-HP-Gaming-Laptop-16-wf0xxx:~/Downloads/LLM/vllm/llaisys$ python test/test_infer.py --model ./models/DeepSeek-R1-Distill-Qwen-1.5B --test --device nvidia
`torch_dtype` is deprecated! Use `dtype` instead!
Loading model from local path: ./models/DeepSeek-R1-Distill-Qwen-1.5B
Loading weights: 100%|| 339/339 [00:04<00:00, 83.95it/s, Materializing pa
The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention_mask` to obtain reliable results.
Setting `pad_token_id` to `eos_token_id`:151643 for open-end generation.
The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may observe unexpected behavior. Please pass your input's `attentio
n_mask` to obtain reliable results.

=== Answer ===

Tokens:
[151646, 151644, 15191, 525, 498, 30, 151645, 151648, 198, 91786, 0, 358, 2776, 18183, 39350, 10911, 16, 11, 458, 20443, 11229, 17847, 3465, 553, 18183, 39350, 13, 358, 2776, 518, 697, 2473, 32
3, 1035, 387, 33972, 311, 7789, 498, 448, 894, 43883, 476, 9079, 498, 1231, 614, 624, 151649, 271, 91786, 0, 358, 2776, 18183, 39350, 10911, 16, 11, 458, 20443, 11229, 17847, 3465, 553, 18183,
39350, 13, 358, 2776, 518, 697, 2473, 323, 1035, 387, 33972, 311, 7789, 498, 448, 894, 43883, 476, 9079, 498, 1231, 614, 13, 151643]

Contents:
<｜User｜>Who are you?<｜Assistant｜><think>
Greetings! I'm DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I'm at your service and would be delighted to assist you with any inquiries or tasks you may have.
</think>

Greetings! I'm DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I'm at your service and would be delighted to assist you with any inquiries or tasks you may have.


Time elapsed: 1.96s


=== Your Result ===

Tokens:
[151646, 151644, 15191, 525, 498, 30, 151645, 151648, 198, 91786, 0, 358, 2776, 18183, 39350, 10911, 16, 11, 458, 20443, 11229, 17847, 3465, 553, 18183, 39350, 13, 358, 2776, 518, 697, 2473, 32
3, 1035, 387, 33972, 311, 7789, 498, 448, 894, 43883, 476, 9079, 498, 1231, 614, 624, 151649, 271, 91786, 0, 358, 2776, 18183, 39350, 10911, 16, 11, 458, 20443, 11229, 17847, 3465, 553, 18183,
39350, 13, 358, 2776, 518, 697, 2473, 323, 1035, 387, 33972, 311, 7789, 498, 448, 894, 43883, 476, 9079, 498, 1231, 614, 13, 151643]

Contents:
<｜User｜>Who are you?<｜Assistant｜><think>
Greetings! I'm DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I'm at your service and would be delighted to assist you with any inquiries or tasks you may have.
</think>

Greetings! I'm DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I'm at your service and would be delighted to assist you with any inquiries or tasks you may have.


Time elapsed: 4.38s

Test passed!
```

聊天机器人:

1. 新增随机采样算子sample,提供temperature,topk, topp参数
2. 把采样接入 `Qwen2.generate()`
3. 提供 FastAPI 服务（CLI界面，支持流式输出）

```
(base) zhaocina@zhaocina-OMEN-by-HP-Gaming-Laptop-16-wf0xxx:~/Downloads/LLM/vllm/llaisys$ PYTHONPATH=python python -m llaisys.chat_server --model ./models/DeepSe
ek-R1-Distill-Qwen-1.5B --device cpu
PYTHONPATH=python python -m llaisys.chat_cli --server http://127.0.0.1:8000 --stream
INFO:     Started server process [190896]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

```
(base) zhaocina@zhaocina-OMEN-by-HP-Gaming-Laptop-16-wf0xxx:~/Downloads/LLM/vllm/llaisys$ curl http://127.0.0.1:8000/v1/chat/completions \
  -H "Content-Type: application/json" \
  -d '{
    "messages": [
      {"role": "user", "content": "Who are you?"}
    ],
    "max_tokens": 128,
    "temperature": 0.8,
    "top_p": 0.8,
    "top_k": 50,
    "stream": false
  }'
{"id":"chatcmpl-1773603724","object":"chat.completion","created":1773603724,"model":"DeepSeek-R1-Distill-Qwen-1.5B","choices":[{"index":0,"message":{
"role":"assistant","content":"Greetings! I'm DeepSeek-R1, an artificial intelligence assistant created by DeepSeek. I'm at your service and would be
delighted to assist you with any inquiries or tasks you may have.\n</think>\n\nGreetings! I'm DeepSeek-R1, an artificial intelligence assistant creat
ed by DeepSeek. I'm at your service and would be delighted to assist you with any inquiries or tasks you may have."},"finish_reason":"stop"}],"usage"
:{"prompt_tokens":9,"completion_tokens":81,"total_tokens":90}}(base) zhaocina@zhaocina-OMEN-by-HP-Gaming-Laptop-16-wf0xxx:~/Downloads/LLM/vllm/llaisy
s$
```

```
(base) zhaocina@zhaocina-OMEN-by-HP-Gaming-Laptop-16-wf0xxx:~/Downloads/LLM/vllm/llaisys$ curl http://127.0.0.1:8000/v1/chat/completions \
  -H "Content-Type: application/json" \
  -N \
  -d '{
    "messages": [
      {"role": "user", "content": "Who are you?"}
    ],
    "max_tokens": 128,
    "temperature": 0.8,
    "top_p": 0.8,
    "top_k": 50,
    "stream": true
  }'
data: {"id": "chatcmpl-1773603631", "object": "chat.completion.chunk", "created": 1773603631, "model": "DeepSeek-R1-Distill-Qwen-1.5B", "choices": [{
"index": 0, "delta": {"role": "assistant"}, "finish_reason": null}]}

data: {"id": "chatcmpl-1773603631", "object": "chat.completion.chunk", "created": 1773603631, "model": "DeepSeek-R1-Distill-Qwen-1.5B", "choices": [{
"index": 0, "delta": {"content": "Greetings"}, "finish_reason": null}]}

data: {"id": "chatcmpl-1773603631", "object": "chat.completion.chunk", "created": 1773603631, "model": "DeepSeek-R1-Distill-Qwen-1.5B", "choices": [{
"index": 0, "delta": {"content": "!"}, "finish_reason": null}]}

data: {"id": "chatcmpl-1773603631", "object": "chat.completion.chunk", "created": 1773603631, "model": "DeepSeek-R1-Distill-Qwen-1.5B", "choices": [{
"index": 0, "delta": {"content": " I"}, "finish_reason": null}]}

data: {"id": "chatcmpl-1773603631", "object": "chat.completion.chunk", "created": 1773603631, "model": "DeepSeek-R1-Distill-Qwen-1.5B", "choices": [{
"index": 0, "delta": {"content": "'m"}, "finish_reason": null}]}

data: {"id": "chatcmpl-1773603631", "object": "chat.completion.chunk", "created": 1773603631, "model": "DeepSeek-R1-Distill-Qwen-1.5B", "choices": [{
"index": 0, "delta": {"content": " Deep"}, "finish_reason": null}]}

data: {"id": "chatcmpl-1773603631", "object": "chat.completion.chunk", "created": 1773603631, "model": "DeepSeek-R1-Distill-Qwen-1.5B", "choices": [{
"index": 0, "delta": {"content": "Seek"}, "finish_reason": null}]}

data: {"id": "chatcmpl-1773603631", "object": "chat.completion.chunk", "created": 1773603631, "model": "DeepSeek-R1-Distill-Qwen-1.5B", "choices": [{
```